

Sasan Liaw

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Education

University of California, Berkeley

Graduating May 2028

Bachelor of Science in Electrical Engineering and Computer Sciences

GPA: 3.67

Coursework: Linear ICs, Microelectronic Devices and Circuits, Signals and Systems, Computer Architecture, Microelectromechanical Systems. In Progress: Digital Design and ICs, Application-Specific ICs.

Technical Skills

Design and Analysis: Cadence Virtuoso, Altium Designer, KiCad, Ansys HFSS, LTspice, Autodesk Fusion 360.

Programming: Java, Python, Git, RISC-V. Familiar with C, C++, SQL, LaTeX, Scheme.

Fabrication & Testing: Soldering, Solder Reflow, Electronic Testing Equipment (Oscilloscope, Adjustable Power Supply, Function Generator, and WaveForms Network Analyzer), 3D Printing, Laser Cutting.

Experience

Space Enterprise at Berkeley (Liquid Rocketry Club)

Berkeley, CA

Avionics Engineer and Mentor

September 2024 - Present

- Design PCBs that process sensor data, manage power, and interface with the engine test stand architecture, including a 16-channel thermocouple amplifier and a scalable Hall effect-based fuel level tracker.
- Lead a semesterly 8-week class on circuit analysis, schematic capture, and layout, guiding students through their first PCB design project from component choice to manufacturing and assembly.
- Integrated wire harnesses, debugged DAQ PCBs, and analyzed telemetry over 500+ hours of onsite testing.
- Wrote custom firmware that executed automated calibration procedures and decoded raw sensor data.
- Collaborated with mechanical teams to seamlessly integrate sensor modules into the liquid propulsion system.

Marvell Nanofabrication Laboratory

Berkeley, CA

Process Engineering Intern

January 2026 - September 2026

- Developed micron and sub-micron scale photolithography process specifications, achieving optimal feature resolution on SiO₂ and metal thin films (Al, W, Ti) using i-line photoresists (MiR701, 12XT).
- Quantitatively verified photoresist exposure quality through optical and scanning electron microscopy.

Theoretical and Applied Fluid Dynamics Laboratory (TAFLab)

Berkeley, CA

Undergraduate Researcher

August 2025 - July 2026

- Designed PCBs that manage power distribution and data acquisition for autonomous ocean-monitoring drones.
- Characterized solar panel power generation, validating its ability to sustain the system's average power draw.
- Implemented multi-rail power distribution (dropping 12V to 8V, 5V, and 3.3V) using LM2596 buck converters, optimizing layout by minimizing switching loops, isolating feedback paths, and utilizing large copper planes.
- Designed for testability, strategically placing test points, indicator LEDs, and sensor bypass pins.

Projects

Display Driver Operational Amplifier

April - May 2026

- Designed a telescopic differential input amplifier with a Class AB output stage to drive a 272×340 LCD at 60Hz using a single voltage supply and current reference, achieving 1.4V output swing in 120ns with <0.2% error.
- Validated the design in Cadence Virtuoso, iteratively improving power consumption, CMRR, and phase margin.
- Balanced trade-offs to meet all specifications, leveraging custom scripts to calculate transistor sizing and biasing conditions using gm/Id design methodology.

Magnetic Fuel Level Sensing

October 2024 - Present

- Engineered and fabricated a modular tracking device using Hall effect sensors, 8-channel ADCs, passive filtering, and a SPI bus, achieving ±0.25 cm positional accuracy with scalable architecture.
- Designed the schematic and layout of a sensor module that can interface with identical boards to create a sensor array of adjustable length while meeting a strict 11 mm maximum thickness constraint.
- Improving the device's reliability as a fuel level tracker following its integration into a rocket propellant tank.